



CIE A Level Chemistry



Your notes

4.1 The Gaseous State: Ideal & Real Gases & $pV = nRT$

Contents

- * Gas Pressure
- * Ideal Gases



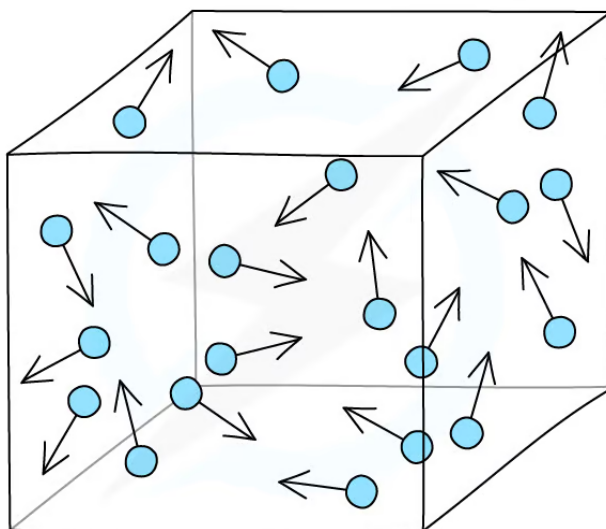
Your notes

Gas Pressure

Gas Pressure

- **Gases** in a container exert a **pressure** as the gas molecules are constantly **colliding** with the wall of the container

Illustration of gas pressure



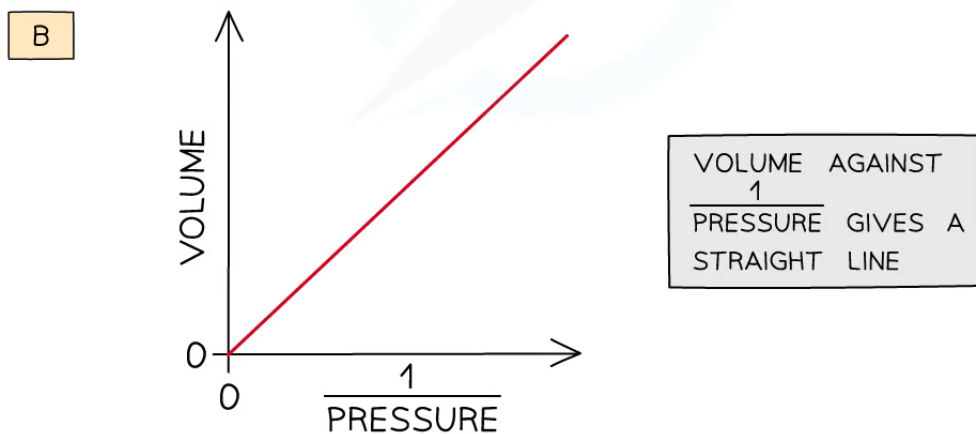
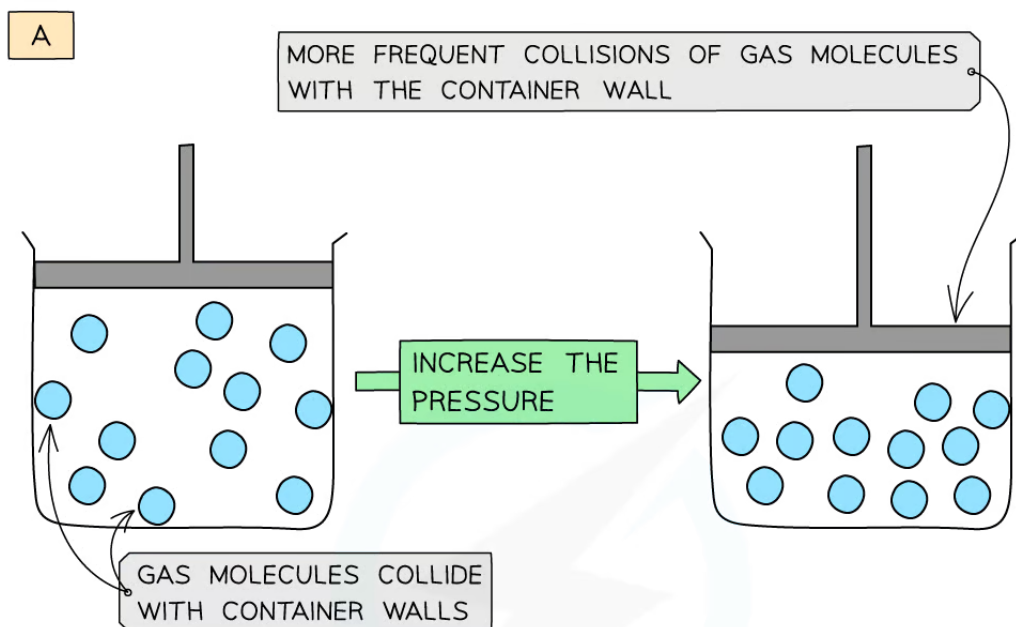
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Gas particles exert a pressure by constantly colliding with the walls of the container

Changing gas volume

- **Decreasing** the **volume** (at constant temperature) of the container causes the molecules to be **squashed** together which results in more **frequent** collisions with the container wall
- The **pressure** of the gas **increases**
- The **volume** is therefore **inversely proportional** to the **pressure** (at constant temperature)
 - A graph of the **volume** of gas plotted against $1/\text{pressure}$ gives a straight line

How decreasing the volume of a gas affects collision frequency



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Decreasing the volume of a gas causes an increased collision frequency of the gas particles with the container wall (a); volume is inversely proportional to the pressure (b)

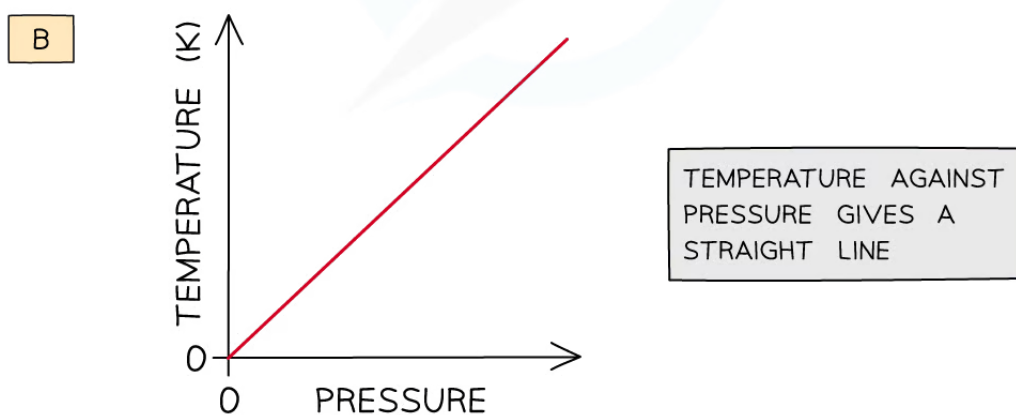
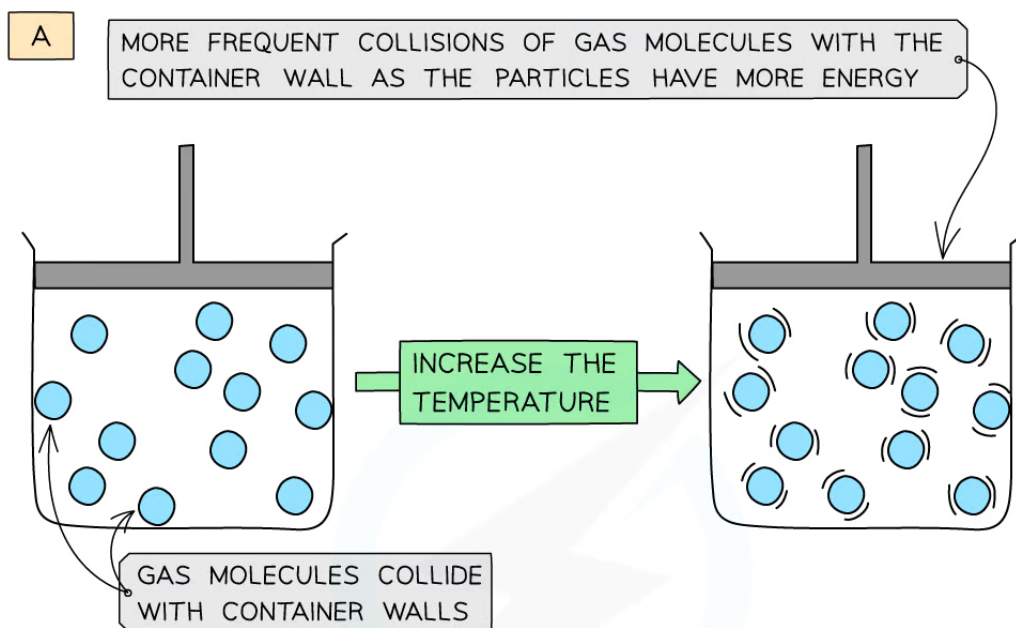
Changing gas temperature

- Increasing the **temperature** (at constant volume) of the gas causes the molecules to gain more **kinetic energy**
- This means that the particles will move **faster** and **collide** with the container walls more **frequently**

- The **pressure** of the gas **increases**
- The **temperature** is therefore **directly proportional** to the **pressure** (at constant volume)
 - A graph of the **temperature** of gas plotted against **pressure** gives a straight line



How increasing the temperature of a gas affects collision frequency



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Increasing the temperature of a gas causes an increased collision frequency of the gas particles with the container wall (a); temperature is directly proportional to the pressure (b)



Your notes

Ideal Gases

Ideal Gas Law

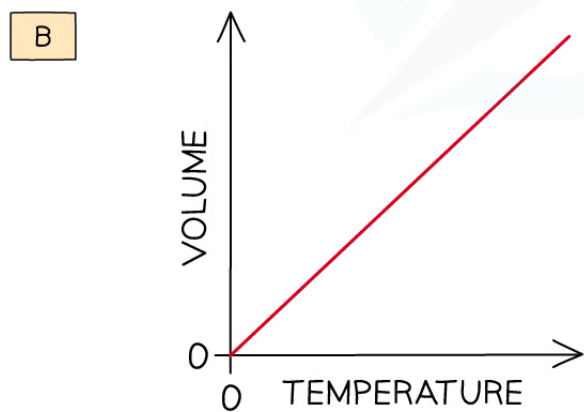
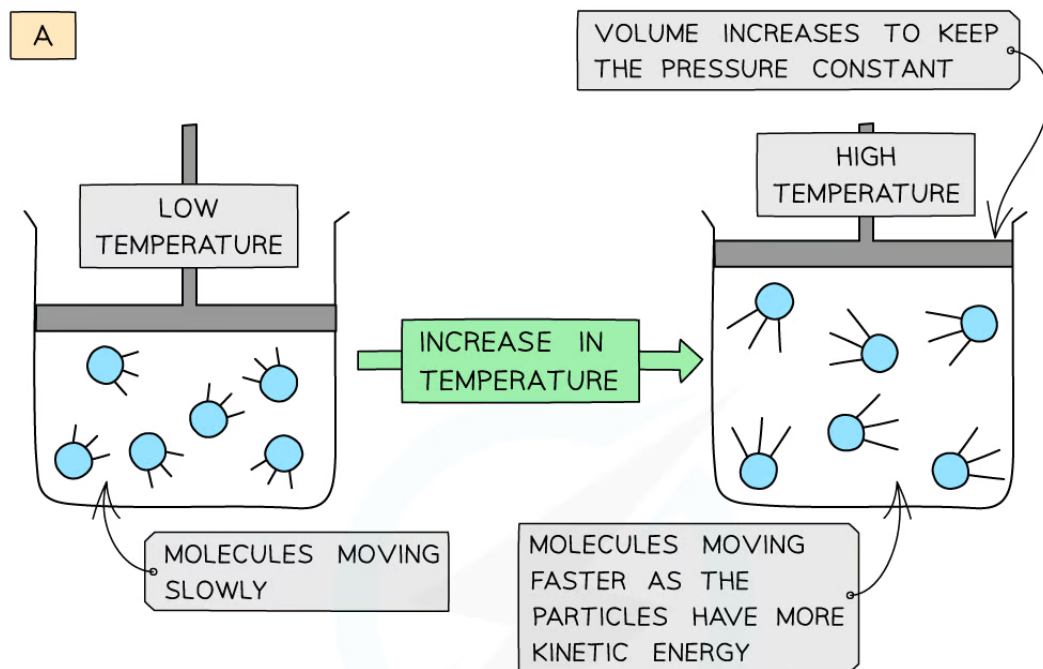
Kinetic theory of gases

- The **kinetic theory of gases** states that molecules in gases are constantly moving
- The theory makes the following assumptions:
 - The gas molecules are moving very fast and randomly
 - The molecules hardly have any volume
 - The gas molecules do not attract or repel each other (**no intermolecular forces**)
 - No kinetic energy is lost when the gas molecules collide with each other (**elastic collisions**)
 - The temperature of the gas is related to the average kinetic energy of the molecules
- Gases that follow the kinetic theory of gases are called **ideal gases**
- However, in reality, gases do not fit this description exactly **but** may come very close and are called **real gases**

Ideal gases

- The volume that an ideal gas occupies depends on:
 - Its pressure
 - Its temperature
 - See section **Changing gas volume** on 1.4.1 Gas Pressure
- When a gas is **heated** (at constant pressure) the particles gain more **kinetic energy** and undergo more **frequent collisions** with the container wall
- To keep the **pressure constant**, the molecules must get further apart and therefore the **volume increases**
- The **volume** is therefore **directly proportional** to the **temperature** (at constant pressure)

How the volume of a gas increases upon heating



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The volume of a gas increases upon heating to keep a constant pressure (a); volume is directly proportional to the temperature (b)

Limitations of the ideal gas law

- At very **high pressures** and **low temperatures** real gases do not obey the kinetic theory as under these conditions:
 - Molecules are close to each other

- There are instantaneous dipole-induced dipole or permanent dipole-permanent dipole forces between the molecules
 - These attractive forces pull the molecules away from the container wall
 - The volume of the molecules is not negligible
- Real gases therefore do not obey the following kinetic theory **assumptions** at high temperatures and pressures:
- There is zero attraction between molecules (due to attractive forces, the **pressure** is **lower** than expected for an ideal gas)
 - The volume of the gas molecules can be ignored (**volume** of the gas is **smaller** than expected for an ideal gas)



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Ideal Gas Equation

Ideal gas equation

- The **ideal gas equation** shows the relationship between pressure, volume, temperature and number of moles of gas of an ideal gas:

$$pV = nRT$$

p = pressure (pascals, Pa)

V = volume (m^3)

n = number of moles of gas (mol)

R = gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

T = temperature (kelvin, K)

- The ideal gas equation can also be used to calculate the **molar mass** (M_r) of a gas

Worked example

Calculating the volume of a gas

Calculate the volume occupied by 0.781 mol of oxygen at a pressure of 220 kPa and a temperature of 21°C .

Answer

- Step 1:** Rearrange the ideal gas equation to find the volume of gas:

$$V = \frac{nRT}{p}$$

- Step 2:** Check that values have the correct units:

- $p = 220 \text{ kPa} = 220\,000 \text{ Pa}$

- $n = 0.781 \text{ mol}$

- $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

- $T = 21^\circ\text{C} = 294 \text{ K}$

- Step 2:** Calculate the volume the oxygen gas occupies:

$$V = \frac{0.781 \text{ mol} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 294 \text{ K}}{220\,000 \text{ Pa}}$$

- $V = 0.00867 \text{ m}^3$

- $V = 8.67 \text{ dm}^3$



Worked example

Calculating the molar mass of a gas

A flask of volume 1000 cm^3 contains 6.39 g of a gas. The pressure in the flask was 300 kPa and the temperature was 23°C .

Calculate the relative molecular mass of the gas.

Answer

- **Step 1:** Rearrange the ideal gas equation to find the number of moles of gas:

$$n = \frac{pV}{RT}$$

- **Step 2:** Check that values have the correct units:

- $p = 300 \text{ kPa} = 300\,000 \text{ Pa}$

- $V = 1000 \text{ cm}^3 = 0.001 \text{ m}^3$

- $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

- $T = 23^\circ\text{C} = 296 \text{ K}$

- **Step 3:** Calculate the number of moles of gas:

- $$n = \frac{300\,000 \text{ Pa} \times 0.001 \text{ m}^3}{8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 296 \text{ K}}$$

- $n = 0.12 \text{ mol}$

- **Step 4:** Calculate the molar mass using the number of moles of gas:

- $$n = \frac{\text{mass}}{\text{molar mass}}$$

- $$\text{molar mass} = \frac{6.39}{0.12 \text{ mol}}$$

- $$\text{molar mass} = 53.25 \text{ g mol}^{-1}$$

Exam Tip

Ideal gases have **zero particle volume** (the particles are really small) and **no intermolecular forces of attraction or repulsion**.

To calculate the temperature in **Kelvin**, add 273 to the Celsius temperature, eg. 100°C is 373 Kelvin.

Remember: an **ideal gas** will have a **volume** that is **directly proportional** to the **temperature** and **inversely proportional** to the **pressure**.